

§ 2.8 负温度状态

一、温度

热平衡原理(或热力学第零定律):

热平衡具有可传递性, 称为热平衡原理, 或热力学第零定律.

所有彼此达到热平衡的系统必然有一个物理量是彼此相等的, 这个物理量就称为温度 T .

分子动理论:

温度标志着物体内部大量分子无规则运动的剧烈程度, 分子热运动越剧烈, 其温度就越高, 反之, 温度越低. 所以说温度是大量分子的平均动能的标志.

统计物理: 微观状态数代表系统的热力学几率

$$S = k \ln \Omega, \quad T = \left(\frac{\partial E}{\partial S} \right)_{N,V}, \quad T = \frac{\Omega}{k} \left(\frac{\partial E}{\partial \Omega} \right)_{N,V}$$

$$E = -N\varepsilon \tanh(\beta\varepsilon)$$

$$S = Nk \left\{ \ln 2 + \ln [\cosh(\beta\varepsilon)] - (\beta\varepsilon) \tanh(\beta\varepsilon) \right\}$$

$$T = \left(\frac{\partial E}{\partial S} \right)_{N,V}$$

粒子数和能量守恒: $N_1 + N_2 = N, \quad E = (N_1 - N_2)\varepsilon$

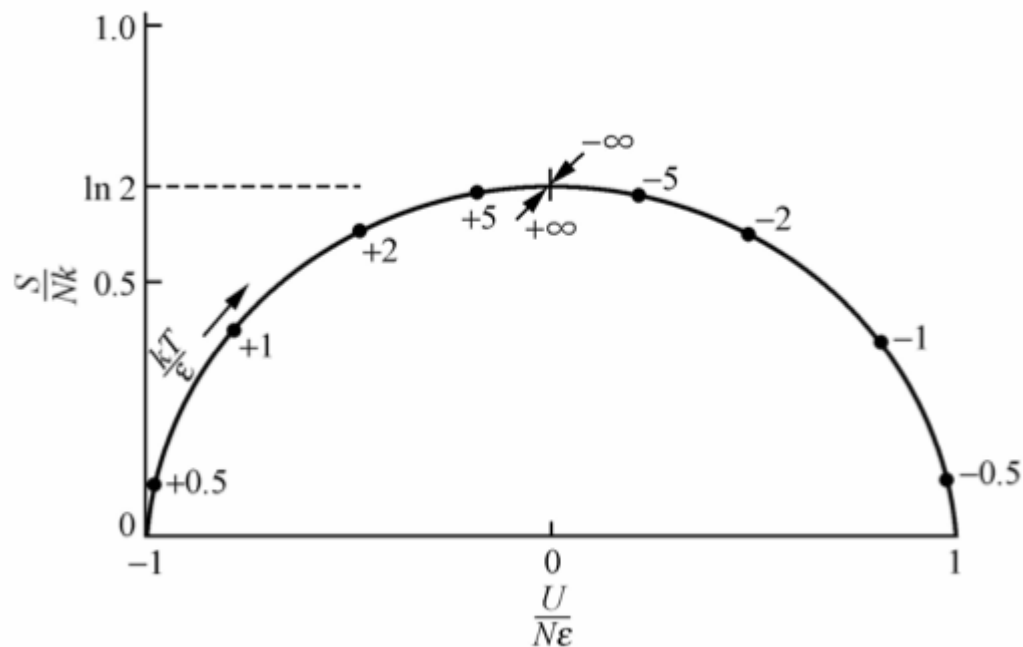
$$N_1 = \frac{N}{2} \left(1 + \frac{E}{N\varepsilon} \right), \quad N_2 = \frac{N}{2} \left(1 - \frac{E}{N\varepsilon} \right)$$

熵: $S = k \ln \Omega = k \ln \frac{N!}{N_1! N_2!}$

$$= Nk \left[\ln 2 - \frac{1}{2} \left(1 + \frac{E}{N\varepsilon} \right) \ln \left(1 + \frac{E}{N\varepsilon} \right) - \frac{1}{2} \left(1 - \frac{E}{N\varepsilon} \right) \ln \left(1 - \frac{E}{N\varepsilon} \right) \right]$$

温度:

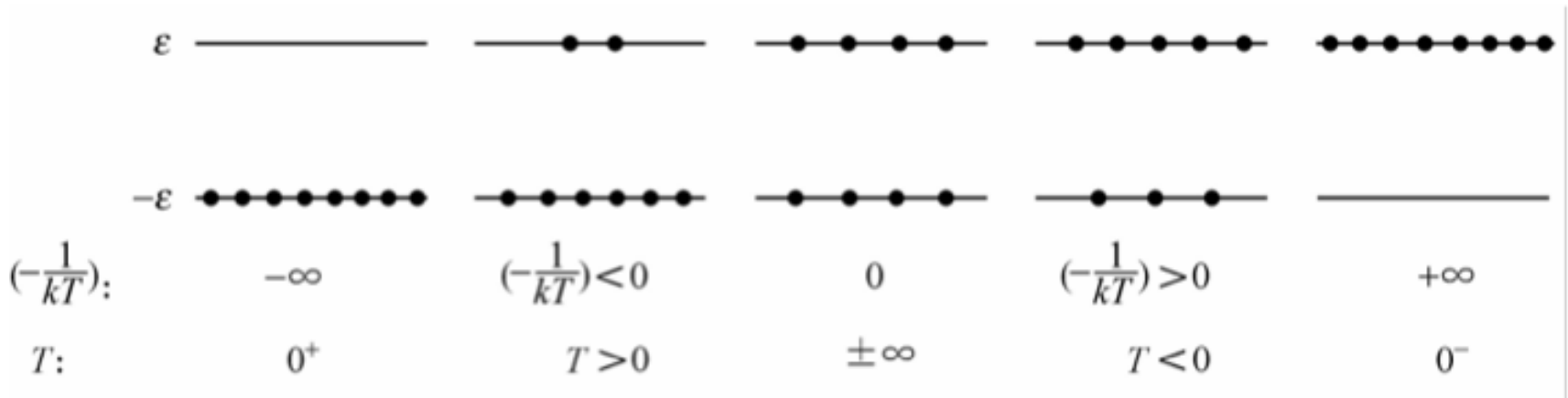
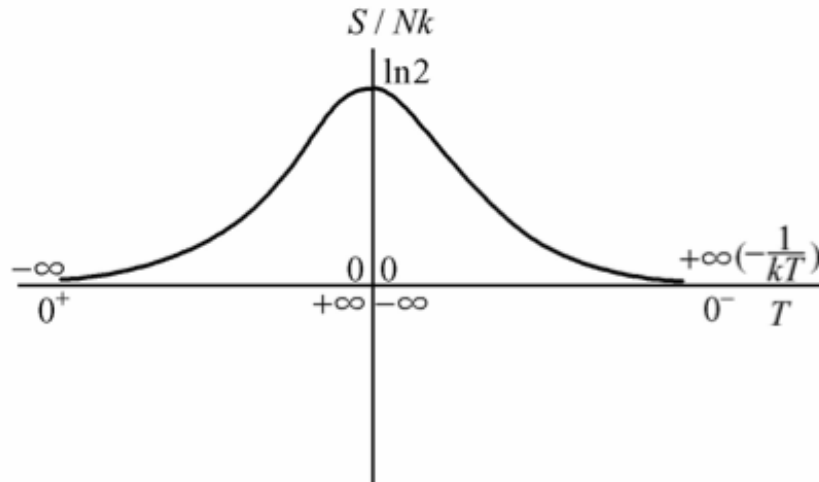
$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right) = \frac{k}{2\varepsilon} \ln \frac{N\varepsilon - E}{N\varepsilon + E}$$



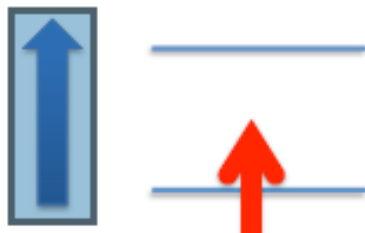
- $T=0$: 磁矩平行, 系统的能量为最小, 系统的熵 $S=0$.
- 温度升高, 磁矩开始反向, 系统内能与熵逐渐增加.
- $T = +\infty$ 时, 正反向概率相等, 熵为最大值.
- 温度继续升高, 能量增加的同时, 系统微观状态数却反而减少, 熵减小, 故处于负温度状态.

$$S = Nk \left\{ \ln 2 + \ln \left[\cosh \left(\frac{\mu B}{kT} \right) \right] - \left(\frac{\mu B}{kT} \right) \tanh \left(\frac{\mu B}{kT} \right) \right\}$$

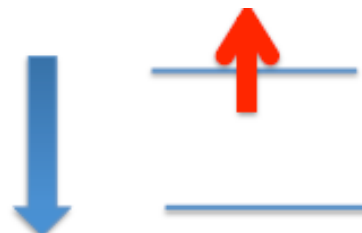
$$E = -N\varepsilon \tanh(\beta\varepsilon)$$



基态、正温度



激发态、负温度



实现负温度的条件：

粒子的能级存在上限；

系统内部弛豫时间足够短；

负温度系统必须与正温度系统隔绝

实验验证：对纯氟化锂晶体中核自旋系统性质的实验研究中1951年玻色耳 (Purcell) 和庞德 (Pound) 将LiF晶体置于强磁场下, 让磁场迅速反向, 使得自旋来不及反向, 在短时间里就实现了核自旋粒子数反转, 从而实现了负温度状态。

A Nuclear Spin System at Negative Temperature

E. M. PURCELL AND R. V. POUND

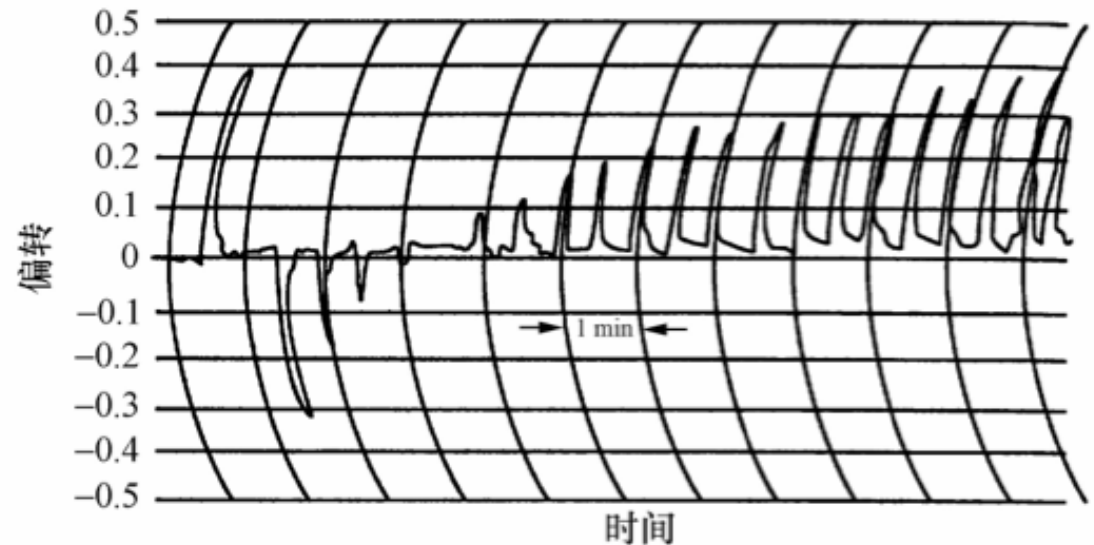
Department of Physics, Harvard University, Cambridge, Massachusetts

November 1, 1950

$$\mu_e = \frac{e\hbar}{2m}$$

$$\mu_N = \frac{e\hbar}{2M}$$

$$\mu B \ll kT$$



Edward Mills Purcell
1912-1997



The Nobel Prize in Physics 1952 jointly with Felix Bloch *"for their development Of new methods for nuclear magnetic Precision measurements and discoveries In connection therewith"*.

Norman Foster Ramsey, Jr.
1915-2011



***The Nobel Prize in Physics 1989 "for the Invention of the separated oscillatory Fields method and its use in the hydrogen Maser and other atomic clocks"*.**